Edible fungi quality and safety traceability management system

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***Abstract*-In recent years, food safety incidents have occurred continuously, and quality safety traceability has become an important means to ensure food safety. This paper analyzes the industrial production process of edible fungi, and the quality and safety traceability management system of edible fungi based on Web was established. The system uses ASP.NET , EAN/UCC-128 coding and XML data exchange technology to realize the quality control of edible fungi production process. Consumers can query the source information of edible mushroom products through this system. Such as edible fungus production information and processing information. The system is also helpful to the risk management and control of edible fungus production enterprises, to minimize the risk in the production process. The practice results show that, the system can meet the whole process quality tracking requirements of edible fungi products.**

Keywords-edible fungi, food safety, coding, data exchange

I. INTRODUCTION

The industrialized production of edible fungi is the product of the combination of traditional agriculture and modern industrial production. With China's entry into WTO, the competition in the international edible fungi market is becoming increasingly fierce, and the technical barriers to food safety in foreign countries have increased significantly. Chinese edible fungi have been rejected, detained, returned, claimed for compensation, terminated the contract, and stopped trade exchanges from time to time because of heavy metals, formaldehyde, sulfur dioxide, etc. exceeding the standards of the importing countries. It is urgent to establish a set of effective tracking system and "rapid warning system" for the whole process of edible fungi quality and safety supply chain.

At present, the internationally used methods for product quality control include: HACCP, GMP and ISO 9000. However, both HACCP and GMP mainly control the single process, lacking effective means to link the whole supply chain.

Aiming at the actual situation that the industrialized production equipment and technology of edible fungi are backward, the management level is low, and the production quality of edible fungi is not high, this paper uses EAN.UCC bar code technology establishes a traceability management system based on distributed application mode, in order to strengthen the quality and safety management of edible fungi, minimize the occurrence of quality and safety accidents, and improve the utilization of resources and the enthusiasm of operators. The establishment and application of the system is of great significance to the formation of standardized and standardized operation management and traceability of edible fungi quality, to improve the management level of industrialized production of edible fungi, to improve the quality of edible fungi products and to ensure food safety.

II. SYSTEM FUNCTIONS

The purpose of establishing food safety information management system is to achieve effective tracking and traceability. The basic production process of edible fungi includes inoculation, fungus generation, fungus scratching, growth, collection, packaging, warehousing, etc. Based on the full investigation of the industrialized production process of edible fungi, this paper designs the basic functional modules of the system, including basic information management, real-time monitoring of mushroom room environment, production activity management, quality and safety management, inventory management and traceability query. A unique barcode traceability number is generated by batch, which registers the production, processing and outbound sales information of edible fungi, and realizes the barcode generation, printing, decoding and traceability of edible fungi products. Release authoritative quality and safety information of inspection and quarantine departments on the basis of enterprise archive data. Combined with short message module, intelligent environment early warning is realized. Consumers can trace the product production process according to the traceability code on relevant traceability websites.

III. SYSTEM DESIGN AND DEVELOPMENT

A. System function design

(1) Basic information management module

Manage various basic data in the production process, for example: the drug information includes the name of the manufacturer, drug name, drug type, specification, etc.

(2) Real time monitoring of mushroom house environment

Through the video monitoring and intelligent environment early warning module, managers can remotely monitor the situation in the mushroom room, and store the environmental parameters collected by the sensors into the

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database. The real-time environmental monitoring information combined with historical environmental information can judge the overall environment for managers' reference.

(3) Production activity management

In the production process, the system records the basic information of each batch of edible fungi (including strain source, variety and strain, inoculation date, batch quantity, mushroom date, etc.) and the production process information (including medication records, inspection information, room transfer information, etc.), through the barcode batch method, and provides the information to the traceability and safety supervision platform for traceability, inspection and supervision. Establish the archive information of each batch of edible fungi, carry out daily production reports for each batch of edible fungi, and record the inspection, medication and other information.

(4) Inventory management

Warehouse is the last link of edible fungi production. Scientific and effective product management can reduce the operation cost to the greatest extent. This module adopts the purchase, sales and inventory management mode, which can quickly record the information of inbound products, daily outbound information, automatic analysis of inventory, automatic cost accounting, etc.

(5) Retrospective query

After purchasing products, consumers can log in to the website and input the traceability bar code on the label pasted on the package into the specified query box to see a series of corresponding information of the purchased products, which can be seen at a glance from production to processing and packaging, including the raw materials, drugs and inspection information used in the production of the edible fungi. At the same time, you can see the specific mushroom room information, processing workshop information, dealer information, etc.

(6) Quality safety management

The quality and safety management module is a comprehensive management module designed for leaders. Through the statistical analysis of this module, you can comprehensively master and understand the operation status of each link of the enterprise. The system can form multi angle and multi-level analysis through various forms of reports, analysis graphics, early warning tips, etc., so as to improve the comprehensive management ability and rapid decision-making ability of the enterprise. It mainly includes: production summary analysis, PEST analysis, production and processing information analysis, etc.

B. System platform architecture

Considering the system maintenance and workload, the system adopts a three-tier solution of b/s structure, namely browser and server structure. In this structure, the user interface is implemented through the browser, a small part of the transaction logic is implemented at the front end, and the main transaction logic is implemented at the server end. According to the demand analysis of the food safety information management system, the server mainly solves the uploading and storage of data, and the statistical analysis of collected data. The server side is responsible for the management and information release of the Web site.

The presentation layer is used for user interaction, receiving user input, displaying data. The business logic layer is used to ensure the effective operation of the program. The data access layer is used to interact with the database, such as adding, modifying and deleting data. This paper is based on ASP.Net platform deployment three-tier structure, background code language using C#. The business logic layer and data access layer are realized by combining component technology with functional modules. The database adopts SQL Server database.

IV. KEY TECHNOLOGIES

A. Traceability coding

Universal and reasonable coding for products is an effective way to ensure food safety traceability. The food safety system applies the international GS1 identification specification to uniquely code the traceability target objects in the world. The application of the standard enhances the reliability of food sources and the speed of information transmission and processing, and provides accurate information guarantee for food safety supervision. GS1 is developed on the basis of commodity bar code, internationally known as GS1 system, which is composed of coding system, data carrier and electronic data exchange standard protocol.

(1) Coding system

The coding system is the core part, which realizes the unique coding of different items. The system adopts EAN/UCC-128 international coding system. As a global standard system, EAN/UCC-128 is universal and extensible. It is widely used for product identification in food safety traceability system. The coding scheme is shown in Figure 1. The code consists of product identification code (including manufacturer identification code and product code), production date code and production site code. In order to keep the design consistent with the national standards, the product identification code adopts the 14 digit commodity barcode. The prefix code of "country or region code" in China is 690-693. In principle, the enterprise needs to have a manufacturer identification code registered in the China article coding center. The format of production date code is YYMMDD, for example, June 1st, 2021, the format is 210601. The production site code is used to represent the number of the culture site of edible fungi, and its coding rules are determined by the manufacturer.

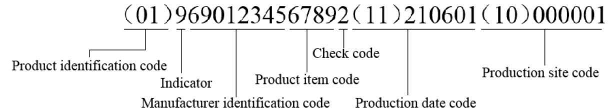

Figure 1. Coding scheme

(2) Data carrier

The data carrier is a carrier that converts the code readable by the naked eye into the code readable by the machine. Bar code labels are easy to make, with high reading accuracy (bit error rate is one millionth) and fast scanning speed. Various bar code generation and collection equipment have mature technology, durability and low price. Considering the promotion of the system and the special advantages of bar code, the system uses bar code as the traceability carrier. The barcode corresponding to the coding scheme is shown in Figure 2.

(01)96901234567892(11)120601(10)000001

Figure 2. Bar code

(3) Data encryption

At present, for different coding systems, there are some problems in the actual application, such as long length, weak encryption or even no encryption, which are easy to be tampered with, forged and unable to achieve the uniqueness of the traceability code. This system uses the coding method of anti-counterfeiting code for reference. Anti-counterfeiting codes are generally divided into fixed codes, random codes, database codes and encryption algorithm codes. The system is designed to paste a traceability mark before each product leaves the factory, which implies a group of unique traceability codes composed of 0-9 digits corresponding to the product. The traceability code is coded according to the coding rules of the system and then transformed and combined again. The transformed traceability code becomes an irregular and disordered code, which makes the forger unable to find its corresponding meaning, so as to effectively prevent the possibility of mass forgery production.

B. XML data exchange technology

In order to realize the forward tracking and reverse tracking of products, one of the important objectives of food safety management system is to realize the unification of product production flow and information flow. However, the internal information structure of enterprises is extremely complex, involving products, design, production, resources, organization, management and other aspects. It is characterized by not only a large amount of data, complex data types and structures, but also complex semantic connections between the data. The data carrier is also multi-media, including a large number of multimedia data such as documents, images and sounds. The advantage of XML data is that it can exchange with the data on the Internet, realize the seamless connection between various data formats, and has a strong openness. XML has the ability to describe all known and unknown data, and has become the basis of data exchange between different systems. In ASP.NET, XML exists on the server side in the form of data resources, and can also be used as a data exchange language between the server side and the client side. In Net framework, the XmlDocument class is used to implement the application interface to read and write XML data.

C. Database

Database design should comply with consistency, integrity, security and scalability, and pay attention to data redundancy. The database constructed in this paper adopts SQL server relational database, establishes different database files for different management sub modules, and takes into account the data sharing among the sub modules. From the perspective of information flow, the system includes data inflow, data storage and data outflow. The efficiency of data query and database security are related to the performance of the whole system.

(1) Database optimization

The system involves a huge amount of data. How to quickly query, analyze and count data from massive data has become one of the key technologies. In the system optimization process, in order to improve the speed of data processing, SQL Server database is selected. Preprocess the data, optimize the query statements, use the statements recognized by the optimizer, make full use of the index, try to avoid the occurrence of table search, reduce the I/O times of table scanning, and improve the system performance and query processing speed. In addition, the data can be partitioned to optimize the resource configuration of the database layer, the flow control of the network layer and the overall design of the operating system layer.

(2) Data security

There are many links involved in the industrialized production of edible fungi. After the system adopts the distributed structure, each link involves the data exchange with the system. The database stores all the information of edible fungi from inoculation to ex warehouse. The possibility of illegal tampering, leakage or destruction of the key data of the system is greatly increased. In order to ensure the confidentiality, integrity and availability of data, the system adopts two ways to strengthen the security protection of system data. On the one hand, the system defines and classifies the key data, and adopts corresponding security technologies for the attributes of each level of key data. Whether querying or other operations, you need to pass validation when accessing the database. On the other hand, role authorization is used to restrict the access rights of system users. The system user login adopts password encryption, and the authorization and security access control strategy of the system is designed through the role authorization and access control model based on group. In addition, in order to ensure the safety of system data, the system database has the function of data backup and restore.

The edible fungus quality and safety management system solution facilitates the capture of data in the process, and the transmission of information is more rapid, accurate and safe, which significantly improves the quality of production operations, realizes the traceability of the whole process, and increases the transparency of supply chain management.

V. CONCLUSION

According to the current situation of edible fungus industry, this paper designs a three-tier distributed structure edible fungus factory production quality and safety management system based on B/S mode. The system provides a set of standard solutions for the identification and traceability of edible fungi products. In view of the data security problems existing in the existing traceability management system, the traceability code encryption and database security processing scheme are proposed, and the system is easy to maintain and upgrade. Basically meet the whole process tracking and tracing of edible fungus products, and establish a food supply chain tracking and tracing system "from farm to table". However, this system is a batch based traceability scheme. The edible fungi of the same batch will be separated and transferred many times in the production process, which brings great difficulties to the refinement of traceability. It will be the focus of the next step to study the more refined traceability scheme of edible fungi and the seamless connection of information.

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